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**Efficiency of the Merger Ringdown
Consistency Test on binaries black holes
population**

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Abstract

When two black holes are gravitationally bonded they form a binary system called black holes binary. These two compact objects rotate around each other emitting gravitational waves in a process called inspiral phase. According to General Relativity with the emission of gravitational waves the two black holes get closer in order to merge and form a final black hole. The remnant object emits gravitational waves to become a Kerr black hole in a process called *ringdown*. The signal provided by the ringdown may be used in several different ways. According to no-hair theorem, black holes are fully described by two quantities: mass and spin. The gravitational waves signal can be used to infer properties of the black holes like mass and spin, where the final spin is supposed to be a result of the angular rotation of the binary. In this project we focus on another usage of ringdown: testing General Relativity. There are plenty of tests of Einstein's theory based on gravitational waves and in this project we address to a particular kind of test called *Merger - Ringdown Consistency Test*. The consistency tests are based on the comparison between data and prediction of a certain theory: in our case General Relativity. The Merger ringdown test can be considered as a null test where we will produce a variable ϵ that is idealistically zero if General Relativity is true and differs from zero otherwise.

In this project we implement the test and observe its performance on a population of $N = 500$ binary black holes. We chose to analyze a population of stellar mass binary black holes using the features of the third generation interferometer Einstein Telescope, which happens to be suitable for stellar mass-type compact objects. After the implementation on the population, we will also debate on the efficiency of the test evaluating the impact of knowing the initial spins of the binaries. The behavior of the entire population can be stacked in a single probability distribution which can be fitted with a gaussian with mean value $\mu_{meas} = 0.013$ and $\sigma_{meas} = 0.006$ when we include initial spin and with mean value $\mu_{known} = 0.004$ and $\sigma_{known} = 0.002$ when we do not. These results show the impact of a prior knowledge for initial spins. The measure of these quantities improves the accuracy on test by one order of magnitude.

In addition we check the performance of the test by adding simulated deviations from General Relativity, first with a constant $c_0 = -0.1$ and then with a gaussian deviation with μ_{dev} and $\sigma_{dev} = 0.01$, showing the behavior of a probability distribution which refers to the whole population.

In the end we conclude this project with different hints on possible development of this work which include both models related and computational challenges.

1 Introduction

After their discovery in 2015 [1] gravitational waves (GWs) became an important tool to explore the universe. Among several activities, testing General Relativity (GR) through GWs has been an interesting goal for the scientific community since this tool can offer precious information about some of the most mysterious object in the universe: black holes (BHs). According to GR, When two BHs are gravitationally bounded they form a binary black hole (BBH), they can rotate around the centre of mass of the binary (*inspiral phase*) and they can merge in a final BH which oscillates in order to become a Kerr BH (*ringdown phase*) [2]. Both phases are characterized by the emission of GWs which can be detected firstly to infer properties of the binary and the remnant BH, then to test GR.

Among tests of GR there is a category that includes *Consistency tests* which fundamental idea is to compare experimental data to GR previsions. One of the most common consistency test is the *inspiral-merger-ringdown test* (IMR) [3], a null test based on the comparison between inferred data from a BBH, which allow to make previsions on the remnant BH using GR, and data from the ringdown signal. If previsions match the properties inferred from ringdown, GR may be considered correct, otherwise there may be a violation of the theory. The IMR inspired the *Merger-Ringdown consistency test* (MRCT) [4], which uses only the ringdown signal in order to check the possible matching between BBH and residual BH properties, where the MRCT may also provides an alternative when the inspiral part of the signal is not detectable. We comment on the differences between these two tests in Chapter II.

How can we perform a consistency test using only the ringdown signal? The idea in the MRCT is to measure properties from both the progenitor BHs and the remnant BH from the ringdown only, using relations between the inspiral and the ringdown phase extracted from numerical simulations based on GR. We measured the remnant's and the binaries properties from the ringdown signal in order to check the consistency between the two assuming the GR relations. We proceed by defining a control variable which compares binary and remnant's properties and is null if GR is true. Specifically, the MRCT compares the spin of the remnant BH as it is directly measured from the ringdown versus the same spin as it inferred from the amplitudes and phases of excitation of the ringdown. It exploits a direct parametrization of the amplitudes and phases as a function of the initial properties of the binary, which are then converted to the final spin via phenomenological GR relations. The difference between the measured and inferred spin, which will be our control variable, is expected to vanish if GR is true. A proof-of-concept of the MRCT has been already implemented in previous work [4] but now we ask: how effective this test may be on a population of BBH? This thesis work focuses on checking this particular application of the MRCT. Since we do not possess a real catalog we had to simulate one assuming GR is true. The catalog is composed by stellar mass BBHs measured by the *Einstein Telescope* (ET) [5], an earth based third generation interferometer which will be able to detect signal from stellar mass BHs. After the implementation on the population, we explored how not knowing the initial spins of the

BBH impacts on the performances of the test. We check the performances of the test with and without assuming exact knowledge of the progenitor spins, in order to quantify the role of the uncertainty on the progenitor spins in the outcome of the test.

This work is organized as follows: in Chapter I we present an introduction to gravitational waves, to the physical characteristics of the ringdown signal and to gravitational wave detectors. In Chapter II we explain how the test has been implemented on a single event, comment in the assumption we had on the waveform and explore the different approaches we had in order to observe the performance of the test. Finally, in Chapter III we present the results of this analysis on the population of BBH we simulated. In the end we give an outlook on possible developments of this work.

2 Chapter I

2.1 The ringdown waveform

The ringdown is the phase where a BH formed by a merger relaxes in order to become a Kerr BH. This process generates the emission of gravitational waves that can be detected by interferometers and can be used to study these systems and test GR. In order to perform the MRCT we have to understand the features of the signal and how to define it. The emission of the ringdown is well described by a damped sinusoid

$$h(t) = h_+(t) + ih_\times(t) = \frac{M}{r} \sum_{lmn} A_{lmn} S_{lmn}^{(+,\times)}(\iota) e^{-t/\tau_{lmn}} e^{i(2\pi f_{lmn}t + \phi_{lmn})} \quad (1)$$

which comes from the analysis of Einstein's equations for a perturbed black hole [2] and represents the signal in the time domain of the gravitational wave the detector receives after the merge of two BHs.

Equation (1) describes the shape of the GWs emitted by a BBH remnant after the merge, but how do we know when ringdown starts? The idea is that analyzing the ringdown requires a choice of the starting time of the process that produces a trade off: the later one chooses the beginning the more you may consider a weak signal; on the other hand if the beginning of the signal has been chosen too early one may consider a non-perturbative part of the signal that does not represent ringdown.

The ringdown waveform is also expressed as a function of spin-two weighted spheroidal harmonics S_{lmn} , a series of complex basis vectors, which will be substituted by the more suitable spherical harmonics that happen to constitute a good assumption under certain hypothesis. In principle, one should write equation (1) in the source frame of the black hole and thus all the quantities appearing in that expression, (M, ω_{lmn} and τ_{lmn}), should be considered in that frame. However an interferometer detects the signals in its frame where cosmological effects has to be taken into account. The result is that all dimensional quantities should be interpreted as redshifted; that means that the instrument detects a redshifted gravitational waves. The prescription to include cosmological effects is very simple: the distance r should be replaced by the luminosity distance $D_L(z)$, and all quantities with dimensions $[\text{mass}]^p$ should enter the waveforms at the detector multiplied by the factor $(1+z)^p$. So, whenever the source is at cosmological distance, r should be replaced by the luminosity distance $D_L(z)$, M by the redshifted mass $(1+z)M^0$, f_{lmn} by the redshifted frequency $f_{lmn}^0/(1+z)$, and τ_{lmn} by $(1+z)\tau_{lmn}^0$ (where all quantities marked by a superscript 0 are measured in the source frame) [6]. In this project all the quantities will already be considered redshifted since the GR is a scale invariant theory and only detector-frame quantities are directly measured. So there will be no need to take into account these factors.

2.1.1 Polarization of the wave

According to equation (1) the ringdown signal can be expressed as a sum of two components $(+, \times)$ which represent the *polarizations of the wave*. Why the signal is a mixture of these two factors? In general, GWs can be described as ripples of the spacetime fabric which can be generated by several phenomena in our universe. Mathematically gravitational waves are linear small perturbations of the metric which, in the simplest example, may be represented by Minkowski metric

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \quad |h_{\mu\nu}| \ll 1 \quad (2)$$

where $h_{\mu\nu}$ is a tensor named *strain*. The strain represents the perturbation of the metric and it is assumed to be small respect to the metric we are considering. We want to find an equation which expresses the shape of the strain generated by a particular source of mass. In GR, mass sources are represented by the *stress-energy tensor* $T_{\mu\nu}$, a quadridimensional energy density, and the relation between the source and the curvature of the space-time is represented by Einstein equation

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu} \quad (3)$$

where $R_{\mu\nu}$ is the *Ricci tensor* (reduced Riemann tensor) and R is the *Ricci scalar*. Einstein's equations represent a system of 16 equations which reduces to 10 due to symmetries of the tensors involved. These quantities are related to the metric by the following

$$R^\mu_{\nu\rho\sigma} = \partial_\rho\Gamma^\mu_{\nu\sigma} - \partial_\sigma\Gamma^\mu_{\nu\rho} + \Gamma^\mu_{\alpha\rho}\Gamma^\alpha_{\nu\sigma} - \Gamma^\mu_{\alpha\sigma}\Gamma^\alpha_{\nu\rho} \quad (4)$$

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}). \quad (5)$$

where $\Gamma^\rho_{\mu\nu}$ represents the *Christoffel symbols*. All these quantities are related to how the mass-energy distribution warps space-time and determine the curvature of its geometry. Gravitational waves have been obtained by substituting the linear metric in equation (2) in Einstein's equation. In order to simplify the calculations we can introduce a more comfortable notation to express the strain

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h \quad (6)$$

and we can gauge away 4 components from the initial 10 independent components of the strain by applying the *Lorentz gauge*

$$\partial^\mu \bar{h}_{\mu\nu} = 0 \quad (7)$$

which physical content is to assume a conservation law for the 4 separate components of the strain.

We remind that applying a gauge represents a change of reference frame

$$x'^\mu \rightarrow x^\mu + \xi^\mu. \quad (8)$$

In case of GR, there is a strong constrain on applying gauge transformation because they change the metric according to

$$g'_{\mu\nu}(x') = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} g_{\rho\sigma}(x).$$

We refer to the previous equation as the *gauge symmetry* of General Relativity. This equation describes how the metric changes as a consequence of a change in the reference frame. When considering linear perturbation of the metric, like in equation (2), the gauge symmetry may be avoided by assuming $|\xi^\mu| < 1$ in equation (8). This assumption implies that the metric stays constant under a specific change of the reference frame. Thanks to the Lorentz gauge one can obtain the following equation, that is the gravitational waves equation

$$\square \bar{h}_{\mu\nu} = \frac{16\pi G}{c^4} T_{\mu\nu}. \quad (9)$$

This equation can be easily solved under the condition that the source is far from where we want to detect GWs

$$T_{\mu\nu} = 0 \quad (10)$$

$$\square \bar{h}_{\mu\nu} = 0 \quad (11)$$

which solution is a plane wave. A very important aspect we have to underline is that the Lorentz gauge in equation (7) is not a strict condition, but the calculation of the strain can be oversimplified by adding two conditions which constitute the *transverse - traceless gauge* or *TT-gauge*. These conditions are the following

$$h^{0\mu} = 0 \quad h_i^i = 0 \quad (12)$$

and they allow to go from 6 independent components to 2. Equation (11) has a plane wave solution which can be expressed as a function of the two components of the strain we are left with. For a wave which propagates in the *z-axis* direction we have

$$h_{ij}(t, z) = \begin{pmatrix} h_+ & h_\times \\ h_\times & -h_+ \end{pmatrix} \cos[\omega(t - z/c)]. \quad (13)$$

The two components of the strain are named *plus polarization* and *cross polarization* and their effect on an object when a gravitational wave passes through it is respectively a squeeze in the (x, y) plane and the variation of the (x, y) components ratio. Equation (13) arms us with an useful tool to describe gravitational waves which polarization components depend on the gauge transformation we apply on the system, or depend on the reference system we are detecting GWs from. It is very important to underline that GWs detected by an earth based interferometer has not be seen in a TT reference frame, which corresponds of a frame where a mass is in free fall, but in the detector reference frame. The detector reference frame is moving with the earth, for this reason the signal will be influenced by Doppler effect from these motions and even from the expansion of the universe which effects have been already introduced. This is an aspect to

account for when detecting GWs which have to be converted in a TT reference frame in order to simplify their shape and use the two polarization we saw. A complete account of this result can be found in [2].

2.1.2 Quasi Normal Modes

According to GR, equation (1) describes the gravitational waves emitted by a Black Hole which propagates energy in order to become a Kerr BH. In particular, the oscillation of the remnant BH is characterized by complex frequencies

$$\omega_{lmn} = \omega_{lmn}^R + i\omega_{lmn}^I \quad (14)$$

which components are named *quasi normal modes* (QNM) and they influence the space and time propagation of the wave

$$\omega^R = 2\pi f \quad (15)$$

$$\omega^I = \frac{1}{\tau}. \quad (16)$$

The l, m, n values are called angular numbers and they describe the different modes that have been excited by the BH. QNMs come from the numerical solution of Teukolsky equations [7], a set of partial differential equations that describes the perturbation of space-time in the vicinity of a Kerr BH [8]. In many fields of science periodic phenomena may be described by the decomposition of a time function in their harmonics. The l, m, n values represent respectively the harmonics, the degeneration of the harmonics and the overtone index. But why are we interested in QNMs? These complex frequencies have a central role in the description of both the GWs and the BHs. The *no-hair theorem* states that QNMs depend only on mass and adimensional spin of the remnant BH [9] since electric charge is observed to be negligible [10]; for this reason measuring QNMs may allow to infer mass and spin of a BH and test the no-hair theorem itself. We underline that QNMs as derived from the Teukolsky equation only describe the oscillation of BHs, while for other compact binaries (e.g. neutron stars) one has to consider matter effects and the description of the oscillations is more complicated [8]. We will not consider neutron stars in this work Testing no-hair theorem needs measures of several QNMs. The ability to measure different QNMs depends on two main factors: the interferometer sensibility and the astrophysical settings of the system. Earth based interferometer like LIGO have a seismic cutoff f_s which limit them at low frequency, therefore they can detect the fundamental mode $l, m = 2$ as long as the BH mass $M < 1.2 \times 10^4 (\text{Hz}/f_s) M_\odot$ if the BH is non-rotating and $M < 2.7 \times 10^4 (\text{Hz}/f_s) M_\odot$ if the BH is rotating near Kerr limit. To sum up, the characteristics of the interferometer are crucial to detect multiple QNMs in order to test no-hair theorem. Secondly, there is a problem when measuring and choosing the QNMs in order to describe the signal in equation (1): the idea, as we anticipated, is that subdominant modes are excited in particular conditions we called astrophysical settings. Numerical simulations show that subdominant modes excitation for binaries black holes

depend on mass ratio $q = m_1/m_2$ where $m_1 > m_2$: the wider is the mass ratio the lower is the excitation [11]. It means that in order to test the no-hair theorem there are some conditions binary black hole have to stick with. Mass ratio is not the only quantity which plays an important role in the excitation of QNMs. In [11] one of the main hypothesis is that initial BHs have no spin: it means that the spin of the final BH is the residual angular momentum of the progenitor binary. In the case of rotating initial BHs it has been observed in [12] that, by the effects of non spinning progenitors, the allowed values in the amplitude-phase are modified. We underline that since BBH have excited subdominant modes under particular condition, the detection of QNMs may be used as a test to recognize the nature of the binary. A more complete review of QNMs, their properties and the conditions of excitation can be found in [8]. For the sake of this project, it is important to understand that initial settings for a BBH influence the excitation of the different harmonics of the wave allowing us to model the waveform more or less accurately.

2.1.3 Waves in the frequency domain

In this paragraph we discuss the application of equation (1) with reference to its use in the data analysis. Most identification models of the waves refers to waves in the frequency domain and not in the time domain; for this reason, we are interested in explaining how to transform to the frequency domain. First of all, we need to introduce the presence of negative modes $m < 0$, since m is defined as $m \in [-l, l]$. we do include modes with $m < 0$ *explicitly*, but we implicitly reabsorb them into modes with $m > 0$ using the symmetric properties of the quasi-normal modes. Generally speaking, we can include negative modes [13] adding a different amplitude, phase and QNMs

$$\omega'_{lmn} = -\omega_{l-mn} \quad (17)$$

$$\tau'_{lmn} = \tau_{l-mn} \quad (18)$$

$$A'_{lmn} = A_{l-mn} \quad (19)$$

in the form

$$h_+ + ih_\times = \frac{M}{r} \sum_{lmn} [A_{lmn} e^{i(\omega_{lmn}t + \phi_{lmn})} e^{-t/\tau_{lmn}} S_{lmn} + A'_{lmn} e^{i(\omega'_{lmn}t + \phi'_{lmn})} e^{-t/\tau'_{lmn}} S'_{lmn}]$$

which can be written according to the intrinsic symmetries of QNMs

$$\begin{aligned} -\omega_{lmn} &= \omega_{l-mn}, \\ 1/\tau_{lmn} &= 1/\tau_{l-mn} \end{aligned}$$

and obtain the following equation

$$h_+ + ih_\times = \frac{M}{r} \sum_{lmn} [A_{lmn} e^{i(\omega_{lmn}t + \phi_{lmn})} e^{-t/\tau_{lmn}} S_{lmn} + A'_{lmn} e^{i(-\omega_{l-mn}t + \phi'_{lmn})} e^{-t/\tau_{l-mn}} S_{l-mn}^*]$$

According to [6]. This equation can be written also as a sum of the real and imaginary part of the strain

$$h_+ = \frac{M}{r} \mathcal{R} [A_{lmn}^+ e^{i(\omega_{lmn}t + \phi_{lmn}^+)} e^{-t/\tau_{lmn}} S_{lmn}] \quad (20)$$

$$h_\times = \frac{M}{r} \mathcal{I} [A_{lmn}^\times e^{i(\omega_{lmn}t + \phi_{lmn}^\times)} e^{-t/\tau_{lmn}} S_{lmn}] \quad (21)$$

where

$$A_{lmn}^{+, \times} e^{i\phi_{lmn}^{+, \times}} = A_{lmn} e^{i\phi_{lmn}} \pm A'_{lmn} e^{-i\phi'_{lmn}}. \quad (22)$$

After we armed ourselves with equations (20) and (21) in the time domain the last step that allow the transition to the frequency domain is to compute the Fourier transform of the previous equation. The Fourier transform has to be computed in the interval $[-\infty, \infty]$ and the time interval we observe the signal is, idelistically, $[0, \infty]$. In order to compute the Fourier transform we assume the wave to be equal for $t < 0$ i.e. we replace the decay factor with $e^{-|t|/\tau_{lmn}}$; this assumption goes as *the Echeverria-Finn convention (EF)* [14, 15]. This correction generates a factor of $\sqrt{2}$ in amplitude to compensate for the doubling of the power in the reflection symmetry. The result of computing the Fourier transform is

$$\int_{-\infty}^{\infty} e^{i\omega t} \left(e^{\pm i\omega_{lmn}t - |t|/\tau_{lmn}} \right) dt = \frac{2/\tau_{lmn}}{(1/\tau_{lmn})^2 + (\omega \pm \omega_{lmn})^2} = 2b_{\pm}. \quad (23)$$

Then the Fourier transforms of the plus and cross components become

$$\tilde{h}_+ = \frac{M}{r} A_{lmn}^+ \left[e^{i\phi_{lmn}^+} S_{lmn} b_+ + e^{-i\phi_{lmn}^+} S_{lmn} b_- \right] \quad (24)$$

$$\tilde{h}_\times = -i \frac{M}{r} A_{lmn}^\times \left[e^{i\phi_{lmn}^\times} S_{lmn} b_+ - e^{-i\phi_{lmn}^\times} S_{lmn} b_- \right]. \quad (25)$$

We also point out that the time of observation is not infinite in a realistic circumstance. The effect of a limited time interval is the loss of accuracy of the Fourier transform and the strain in the frequency domain. A limited time interval means that the integral of the Fourier transform cannot be done in $[-\infty, \infty]$ and the signal cannot be computed with all the existing frequencies. Another issue that is present when computing the Fourier transform is the limited differential time interval between two detections of the same signal. This constrain determine a limited sampling in the frequency domain.

2.2 Data Analysis techniques

In this section we will face some experimental challenges of the detecting gravitational waves, focusing also on the properties of detectors themselves. Like in many fields of science, when we make some measures there is no possibility to avoid noise generated by instruments or other events. This is also the case for gravitational waves, where astrophysical events from several phenomena have to be taken into account. Besides, this project is based on the features of the third generation earth-based interferometer Einstein Telescope, which will play a crucial role in detecting stellar mass type BHs. Here in this section we approach some experimental features and parameters of detectors one should take account of when studying gravitational waves.

2.2.1 Power spectral density

First of all we want to introduce and debate on the sensitivity of a detector. In particular, how to build a function which quantify the sensitivity and its features. We may start with the simple question: what does ‘the noise of the interferometer’ mean? Gravitational waves are often signals that are buried into noise, which we can distinguish in two types: the first is the noise which come from physical events that interact with the interferometer and the second one depends on the components of the detector. This last source of noise shows up due to physical effects of the interferometer’s materials or the presence of different phases of the signal analysis. Assuming the detector may be modelled as a sequence of *stages* where the signal is computed, detector noise may manifest in each one of these stages. Usually this kind of noise is referred as

$$\tilde{n}_{out}(f) = T(f)\tilde{n}(f)$$

where $T(f)$ represents the *transfer function*, which relates the noise of the previous stage to the following one. If the detector has more stages the total transfer function may be expressed as $T = \prod_i T_i(f)$. This type of noise can be determined modelling the different $\tilde{n}_{out}$ and knowing the transfer function for each stage. Since $\tilde{n}_{out}$ depends on the physical features of the detector is not usually considered as the *detector noise*. For this reason, in our project when we refers to noise we mean the noise from physical external sources and not from building features of the detector. We assume that the signal a detector receives to be

$$s(t) = n(t) + h(t) \tag{26}$$

as a function of time which will be Fourier transformed in the frequency domain in order to be analyzed. $n(t)$ represents the noise that comes from external physical phenomena. What is considered the detector noise is the one in equation (26). So, in the above sense, we take $n(t)$ to be the detector’s noise. If the noise is stationary, as we assume for the moment, the different Fourier components are uncorrelated, and therefore the ensemble average of the Fourier components

of the noise is of the form

$$\langle \tilde{n}^*(f) \tilde{n}(f') \rangle = \delta(f - f') \frac{1}{2} S_n(f) \quad (27)$$

where the above equation defines the function $S(f)$ named *power spectral density* (PSD). We refer as the ensemble average the average over many possible 'realizations' of the system. Since we have only one physical system that is our detector we call realization the measure of the noise over a given time interval T instead of measuring different signals (*Ergodic principle*). The ensemble average is therefore an average over different measures of the signal over the same time interval. This hypothesis needs two considerations: the first one is the frequency resolution we obtain ($\Delta f = 1/T$), the second one is that idealistically we assume the noise to be zero on average ($\langle n(t) \rangle = 0$) if measured for an infinite time interval but the limitness of the time we actually perform the measure allow us to consider noise in a finite time interval as the realization we anticipated. If we consider $f = f'$ in a finite interval equation (27) becomes

$$\langle |\tilde{n}^2(f)| \rangle = \frac{1}{2} S_n(f) T \quad (28)$$

$$\frac{1}{2} S_n(f) = \langle |\tilde{n}^2(f)| \rangle \Delta f. \quad (29)$$

So, the PSD represents the ability of the detector to measure a signal depending on a specific frequency. From equation (29) we can notice that the PSD is greater if the noise is higher respect to a frequency f or if the Δf is higher, which is related to the observational time interval. The factor $1/2$ has been added in order to define the following relations with physical frequencies

$$\begin{aligned} \langle |\tilde{n}^2(t)| \rangle &= \langle |\tilde{n}^2(t=0)| \rangle \\ &= \int_{-\infty}^{\infty} df df' \langle \tilde{n}(f) \tilde{n}(f') \rangle \\ &= \int_{-\infty}^{\infty} df S_n(f) \\ &= \int_0^{\infty} df S_n(f). \end{aligned}$$

Following equation (29) we define the *sensitivity* of the detector as $\sqrt{S_n(f)}$. Note that if the noise increases by a factor λ , $n(t) \rightarrow \lambda n(t)$, then $S_n(f) \rightarrow \lambda^2 S_n(f)$ while the strain sensitivity scales linearly. We underline that equation (27) is not rigorous: for example it is not mandatory that $n(t)$ goes to zero at $t \rightarrow \pm\infty$, for this reason the Fourier transform is not well-defined. A more rigorous approach to the problem can be found in [2]. In this project we will refer to the *average psd* of Einstein Telescope (ET), a third generation earth based interferometer. The PSD of Einstein telescope spans a frequency range from 1Hz to 10kHz with the best strain sensitivity of $\sim 10^{-25}(\text{Hz})^{-1/2}$ in the frequency range of 20-200 Hz. The largest angular frequency which a BH of mass M produces

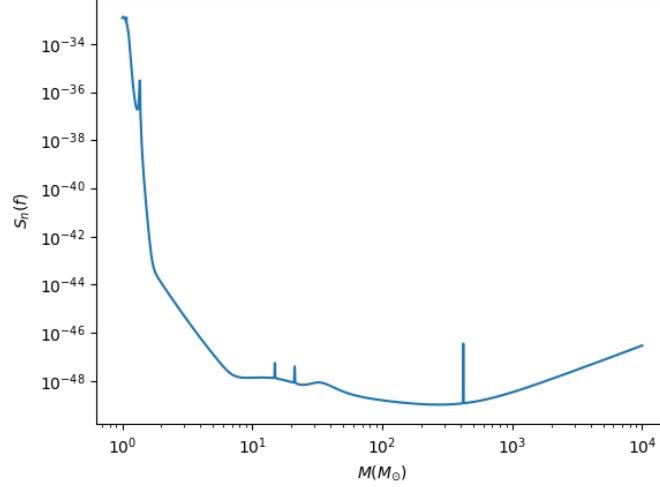


Figure 1: Average PSD of the Einstein Telescope. The frequency range of this interferometer spans in $[1 - 10^4]$ Hz which means a mass range of $[1 - 10^4]M_\odot$

is roughly $\omega^2 \sim M/R^3$ where R is its size. Taking $R = 2M$ a Schwarzschild BH of $10M_\odot$ produces a GW signal with frequency $f = 1.14\text{kHz}(10M_\odot/M)$ and a Kerr BH with spin 0.9 produces a signal with frequency $f = 2.15\text{kHz}(10M_\odot/M)$ [16]. This means that ET's frequency range cover a mass range for BHs of $M \in [1 - 10^4]M_\odot$. For this reason ET happens to be suitable for our project where we analyze the GWs waves signal of stellar mass BHs. Figure 1 shows the PSD of ET. We point out that, as we anticipated, the PSD we consider is an average PSD: a modelled function that have been determined theoretically and does not consider the presence of noise. In a real situation the PSD has to be calculated by subtracting a modelled signal from an actual detection of the interferometer in order to model noise.

2.2.2 Detector Pattern Functions

We learned in the previous section how a gravitational wave can be written as a function of the two polarization h_+ and $h_\times$. We may ask now, what happens when the GW interact with the detector? The main concept we have to present is that detector take scalars as input; so, in order to analyze the signal, detector has to transform the strain tensor in a scalar. This process is allowed by the *detector tensor* D^{ij}

$$h(t) = D^{ij}h_{ij}(t). \quad (30)$$

Introducing the polarization basis e_{ij}

$$e_{ij}^+ = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad e_{ij}^\times = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (31)$$

we can write the strain tensor as a function of these basis in the perpendicular plane respect to the direction of propagation

$$h_{ij}(t) = \sum_{A=+, \times} e_{ij}^A(\hat{n}) h_A(t) \quad (32)$$

where the basis vectors depend on the direction of propagation of the wave. Let's apply the detector tensor in order to obtain a scalar

$$h(t) = \sum_{A=+, \times} D^{ij} e_{ij}^A(\hat{n}) h_A(t). \quad (33)$$

Then we define the *detector pattern functions* $F_A(\hat{n})$

$$F_A(\hat{n}) = D^{ij} e_{ij}^A(\hat{n}). \quad (34)$$

Equation (34) refers to how the detector interact with the gravitational wave depending on the its intrinsic features (contained in D^{ij}) and the direction of propagation of the wave. Pattern functions depend on both the position of the source in the sky, labelled with the variables (θ, ϕ) , and the polarization component. The strain can be written as

$$h(t) = h_+(t) F_+(\theta, \phi) + h_\times(t) F_\times(\theta, \phi). \quad (35)$$

It is interesting to observe that pattern function depends on the reference frame we consider since they depend on the basis vectors e_{ij} . A rotation angle ψ change both basis vectors and the polarization components but not the scalar strain that of course has to be independent from the reference frame [2]. The possible advantage of a choice for the rotation angle is that F_+ or $F_\times$ may vanish in order to simplify the problem. Another feature of interferometers consists in the detection of signal which come from almost all directions, compared to other type of telescope which have to be focused on a specific region of space. A disadvantage is that a single detector cannot determine the position of the source in the sky. A single detector has an output $h(t)$ that, according to eq.(35) depends on four unknown: the two functions $h_{+, \times}(t)$ and angles (θ, ϕ) . It means that only a network of three detectors may solve the problem adding to delayed time for the signal and determine the position of the source.

2.2.3 Matched filtering

In this paragraph we introduce the technique called *matched filtering*. The purpose of matched filtering is to dig out the signal $h(t)$ from noise $n(t)$ when $|h(t)| \ll |n(t)|$. This situation is very common in astrophysics and it can be

solved with a ‘filtering’ procedure. The idea behind this approach is that we can dig out the strain from the noise if we know, at least to some level of accuracy, the form of $h(t)$. Assuming the signal is $s(t) = h(t) + n(t)$ one can multiply the output by $h(t)$ and integrate over observational time T

$$\frac{1}{T} \int_0^\infty dt s(t) h(t) = \frac{1}{T} \int_0^\infty dt h^2(t) + \frac{1}{T} \int_0^\infty dt n(t) h(t). \quad (36)$$

Now, we assume the signal to be a slowly varying function of time

$$\frac{1}{T} \int_0^\infty dt h^2(t) \sim h_0^2 \quad (37)$$

where we approximate the signal contribution with its average. So, the integral in the previous equation grows linearly with T . Since the noise $n(t)$ and $h(t)$ are uncorrelated, the quantity $n(t)h(t)$ is oscillating and its integral will grow only as $T^{1/2}$. The fact they are uncorrelated means that we can separate the integrals where the one with the noise is assumed to be linear with time

$$\frac{1}{T} \int_0^\infty dt n(t) h(t) \sim \left(\frac{\tau_0}{T}\right)^{1/2} n_0 h_0, \quad (38)$$

where n_0 is not an average noise but a characteristic amplitude of the noise and τ_0 is a typical time of an event. We notice that in the limit $T \rightarrow \infty$ the second term cancel out; in this way we ‘filtered’ the noise from the output. Of course, the observable time is not infinite due to the intrinsic duration of the phenomenon and the availability of the interferometer. The focal point here is that there is a sweet spot in the magnitude of the strain we can detect depending on the typical time scale: in order to detect h_0 we must have $h_0 > (\tau_0/T)^{1/2} n_0$ that is a threshold we can’t escape. This problem can be approached in a more rigorous way by defining the *filter function* $K(t)$

$$\hat{s} = \int_{-\infty}^\infty dt s(t) K(t). \quad (39)$$

We now want to find the filter function which correspond to the maximum SNR for such a signal. Since filter function is chosen to ‘match’ the signal that we are looking for, the technique is called matched filtering. The SNR is defined as S/N , where S represents the value of $\hat{s}$ when the signal is present while N is the root mean squared value of $\hat{s}$ when the signal is absent. Since $\langle n(t) \rangle = 0$ we have

$$\begin{aligned} S &= \int_{-\infty}^\infty dt \langle s(t) \rangle K(t) \\ &= \int_{-\infty}^\infty dt h(t) K(t) \\ &= \int_{-\infty}^\infty df \tilde{h}(f) \tilde{K}^*(f) \end{aligned}$$

and

$$\begin{aligned}
N^2 &= [\langle \hat{s}^2(t) \rangle - \langle \hat{s}(t) \rangle^2]_{h=0} \\
&= \langle \hat{s}^2(t) \rangle_{h=0} \\
&= \int_{-\infty}^{\infty} dt dt' K(t) K(t') \langle n(t) n(t') \rangle \\
&= \int_{-\infty}^{\infty} dt dt' K(t) K(t') \int_{-\infty}^{\infty} df df' e^{2\pi i f t - 2\pi i f' t'} \langle \tilde{n}^*(f) \tilde{n}(f') \rangle,
\end{aligned}$$

Therefore

$$\frac{S}{N} = \frac{\int_{-\infty}^{\infty} df \tilde{h}(f) \tilde{K}^*(f)}{[\int_{-\infty}^{\infty} df (1/2) S_n(f) |\tilde{K}(f)|^2]^{1/2}}. \quad (40)$$

In the end we can ask what is the filter function which maximizes S/N for a given $h(t)$. The problem is solved by defining the scalar product between two real functions $A(t)$ and $B(t)$, by

$$(A|B) = Re \int_{-\infty}^{\infty} df \frac{\tilde{A}^*(f) \tilde{B}(f)}{(1/2) S_n(f)} = 4 Re \int_0^{\infty} df \frac{\tilde{A}^*(f) \tilde{B}(f)}{S_n(f)} \quad (41)$$

so that the equation for SNR can be written as

$$\frac{S}{N} = \frac{(u|h)}{(u|u)^{1/2}} \quad (42)$$

with

$$\tilde{u}(f) = \frac{1}{2} S_n(f) \tilde{K}(f). \quad (43)$$

Equation (42) is maximized by a vector u that is parallel with h , so we get

$$\tilde{K}(f) = \text{const.} \frac{\tilde{h}(f)}{S_n(f)} \quad (44)$$

which is the signal itself weighted on the PSD of the detector. Inserting the previous equation into (42) we obtain

$$\frac{S}{N} = (h|h)^{1/2} \quad (45)$$

or

$$\left(\frac{S}{N} \right)^2 = 4 \int_{-\infty}^{\infty} df \frac{|\tilde{h}(f)|^2}{S_n(f)} \quad (46)$$

that is the definition of SNR we already gave. We conclude by underling that matched filters requires the knowledge, or at least an hypotesis, of the GW signal as we saw that this knowledge is necessary to understand the threshold of the strain we can detect among noise and since the signal itself has to be used as filtering function together with the PSD in order to have the maximum SNR. On the other hand, SNR will play a crucial role in the implementation of the test since it represents the relationship between signal and noise at a given frequency. The higher is the SNR the better an interferometer will be able to detect a signal among noise.

3 Chapter II

In this chapter we are going to explore what does it mean to test general relativity with gravitational waves. In particular we will explore the different ideas behind GR tests and focus on a class of test called *consistency test*. Then we will explain the logic the test we implemented is based on and its features.

3.1 Testing General Relativity

We already introduced the opportunity to test GR through gravitational waves and in this paragraph we dig in the landscape of several tests of GR that have been applied to LIGO-VIRGO-KAGRA detections so far. First of all, tests of GR may be divided in 4 classes: the *consistency test* which check the consistency of GR predictions with observed data, the *parametrized tests* which set statistical bounds on the parametrized deviations from GR injecting them in the parameters according to a specific theory of GR, tests based on checking if there are violations of the merger remnant properties assuming GR is true and in the end, *polarization tests* which look for extra polarization modes present in the data. Before describing the ideas behind the different tests and their features, two considerations have to be made when testing GR: firstly one should be careful to consider events with a low *false alarm rate* (FAR) which represents the probability that the event is not generated from an astrophysical source; of course if a violation of GR is detected for an event which is not astrophysical it cannot be considered a violation of GR. Secondly, one has to take into account the systematic errors due to waveform modelling, detector calibration and estimation of the PSD. Another useful consideration that can be made when testing GR is the possibility to inject deviations into data from different events in order to describe the features of the entire population. This process can be made in two ways: *the simple combining approach* which assume a systematic deviation from GR in each event and the *hierarcal combining* which idea is to assume deviations from a probability distribution (usually a Gaussian). After these considerations we proceed by describing the different features of two types of tests we already introduced and we are interested in: consistency and parametrized tests.

3.1.1 Consistency and Parametrized test

The first type of test we introduce is the consistency test. It plays an important role in this project since the test we analyzed belongs to this class. This class of test addresses the simple question: can the observed data be fully explained by assuming GR? There are two types of consistency test based on two different ideas: *residual test* and *inspiral-merger-ringdown consistency test*. Residual test check for signatures left in the data after subtracting the best-fit GR template. So, if GR is the correct theory what is left in the signal has to agree with the instrumental noise. The pipeline for this kind of test is to obtain features of an event through parameters estimation: basically, obtaining those parameters

which maximizes the likelihood of reproducing the detected signal. We stress out the fact that these parameters may not be the most reliable ones since they do not include the presence of a prior assumptions. Once obtained the vector of parameters it can be used to generate the signal observed if GR was correct and then compare it with the detected signal. If the difference does not agree with instrumental noise we may have detected a violation of GR. Of course, the possible error related to the parameters estimation may influence the production of the signal assuming GR is correct. Also, we have by definition that the likelihood is

$$\log \Lambda = -\frac{1}{2} \sum_i ||d_i - h_i|| \quad (47)$$

where the index i represents the number of detectors and d the data. This definition is based on the assumption of gaussian noise n but if there was a violation of GR the difference in the previous equation may not be gaussian. A more quantitative explanation of how this type of test is implemented is in [17].

The second class of consistency test is the inspiral-merger-ringdown (IMR) test. The idea behind this approach is to determine the properties of the remnant BH both from the binary signal and from the remnant signal. The analysis of the binary signal allows to infer the properties of the progenitors in order to calculate the remnant's according to GR. Once these properties (usually mass and spin) have been inferred, the comparison between them determine if GR is true or not. If they match we can consider GR as true. The IMR test takes into account the signal from both inspiral and merge phase separately through the calculation of a threshold in frequency named f_{cut} . f_{cut} represents the spin-dependent innermost stable circular frequency. This threshold has been used to split the signal between the inspiral and the merger part. The purpose of this value is to recognize and divide the signal from different phases of the merge in order to analyze them separately. The progenitors features are inferred by both of inspiral and merge signals and compared in order to verify their agreement if GR is true. Of course these two signals differ for the frequency range they live in, for this reason an high SNR in both regions is needed separately. After this comparison one may use masses and spins of the progenitors from both inspiral and merge phase and obtain an average of these features to consider. These two estimates will be used to calculate the remnant mass and spin according to GR.

In the end we introduce parametrized tests. The principle they are based on is to detect deviations from GR in the generation and in the propagation of GWs. The idea is to add model-independent parametric variations which can be constrained with data. For example, if the wave can be expressed as

$$\tilde{h}(f) = A(f)e^{i\phi(f)} \quad (48)$$

one can add a deviation in the parameter space through

$$\phi_i(f) \rightarrow (1 + \delta\phi_i)\phi_i. \quad (49)$$

These deviations change the shape of the waveform and can be detected with the IMR test comparing both inspiral and ringdown signal or inspiral and merge

models. In fact, the phase these deviations are injected in plays an important role in the analysis. Two approaches have been considered mostly: the first one is based on the addition of the deviations in the inspiral and merge phase of the interaction between BHs and in the second one these additive factors have been applied to the inspiral merge only and they vanish at high frequency, near the merge range [18]. One of the main differences between these two approaches is the dependence on the waveform in the first case that is absent in the second [17].

3.2 Merger Ringdown Consistency test

In this thesis project we consider the merger-ringdown consistency test (MRC) [19]. We derive prospects on its efficiency on a population of BBH and we analyze the impact of measuring progenitors spins in the test. The idea behind MRC is to use the signal from the ringdown phase to infer properties of both the remnant black hole and the progenitors. Then, use the inferred properties of the progenitors to determine the remnant BH they would produce according to GR. In particular, we are interested in inferring the final spin of a BH, the mass ratio and the progenitors spins of the BBH. Finally, a comparison between the measured final spin and an estimate of the same quantity obtained from the mass ratio and from the progenitors spins, may show the presence of a violation of GR. If the final spin inferred directly from ringdown is equal to the final spin we obtain by calculation with inferred mass ratio and initial spins, we may consider GR true. This test has been inspired by IMR but it addresses a different question. While IMR checks for global consistency of the BBH evolution, violation of the merger-ringdown test indicates GR modifications that alter the setup for ringdown. In addition, MRC requires only the ringdown signal; this may be an advantage in a situation where the inspiral part of the signal is not accessible.

MRC is a null test, for this reason it is based on the definition of a quantity which is idealistically zero when GR is true and it is not when GR may be violated. We will discuss the test for a single event at first and then on a simulated population of BBH. In order to simulate the signals from the events we had to define an array of parameters $\Theta = \{M, \chi_f, \iota, \theta, \psi, A_{lmn}, \phi_{lmn}, D_L, \tau_{lmn}, f_{lmn}\}$ for the population binaries. Generating this array required numerical fits from GR; the result is that for a simulated population the null test should detect violation from GR only if injected. In our case, we will explore different approaches, starting from injecting GR in order to find no deviation to injecting both systematic and Gaussian deviations, following both combining and hierarchical approach we introduced in the previous section. We refer to equation (24), (25) and (35) as the ringdown signal in the frequency domain. This is a function of the vector of parameters Θ which number depends on the number of angular modes we decide to consider. Here we explore the implementation and the features of the MRC test, which assumption we do, how we relate ringdown parameters and progenitors parameters in order to have a final parametrization that allow us to run the test.

3.2.1 Waveform modelling

First of all, we considered the waveforms in eq. (24) and (25), in the frequency domain, which are associated to the ringdown. We assumed the BBH we analyze to be non spinning and with quasi circular orbits. We underline that this is assumed only to offer a proof-of-concept, but in reality a deviation from quasi-circularity would cause systematic biases in the mass and spin calculation, which may be interpreted as a violation of GR [20].

We substitute the spin-2 spheroidal harmonics S_{lm} with the simpler spherical harmonics Y_{lm} [21]. This assumption is well justified in the low spin range and it is a working approximation also at higher spin [22]. Spherical harmonics depend on the inclination angle of the remnant BH ι , reducing to zero for all the subdominant modes if $\iota = 0, \pi$. We chose to include the $(l, m, n) = (2, 2, 0)$ dominant mode and the $(3, 3, 0), (2, 1, 0), (4, 4, 0)$ subdominant modes which may be the most excited given the features of our population we will introduce in the next section. Indeed it has been observed that subdominant modes are more excited for mass ratio higher than one [23]. Amplitudes of these modes are reported as a function of symmetric mass ratio $\eta = m_1 m_2 / (m_1 + m_2)^2 = q / (1 + q)^2$ instead of the simple mass ratio q , and the effective spin $\chi_{\text{eff}} = 1/2(\delta\chi_+ + \chi_-)$ where

$$\begin{aligned}\delta &= (1 - 4\eta)^{\frac{1}{2}} \\ \chi_+ &= \frac{m_1 \chi_1 + m_2 \chi_2}{m_1 + m_2} \\ \chi_- &= \frac{m_1 \chi_1 - m_2 \chi_2}{m_1 + m_2}.\end{aligned}$$

As we can grasp from the previous equations the spins and the mass ratio of the progenitors determine the excitation of the subdominant modes with amplitudes

$$A_{210}(\eta) = |0.328(1 - 4\eta)^{1/2} + 0.115(1 - 4\eta) - 0.414\chi_{\text{eff}}| A_{220}(\eta) \quad (50)$$

$$A_{330}(\eta) = |0.572(1 - 4\eta)^{1/2} - 0.144(1 - 4\eta) + 0.035\chi_{\text{eff}}| A_{220}(\eta) \quad (51)$$

$$A_{440}(\eta) = [0.251(1 + 59.773\eta^3 - 16.307\eta^2 - 3\eta) - 0.011\chi_+] A_{220}(\eta) \quad (52)$$

These relations come from phenomenological models that include GR based on fits and numerical simulations [12]. Figure 2 shows how fits in the previous equations behave as a function of mass ratio and for different initial spin. As we can notice from the figure there is a difference in the excitation modes depending on the initial spins which has to be taken into account since our hypothesis is to consider non-spinning binaries.

In addition, we assume the binary to have aligned spin, so we suppress the possible precession effects which influence the modulation of the signal and the excitation of QNMs [24, 25, 26]. Thanks to equations (50-52) we managed to write the modes amplitudes as a function of mass ratio and the progenitors spins, lowering the number of parameters. The same process can be done with

the phase ϕ_{lmn} [12] and the following equations

$$d\phi_{210} = 0.401 + 0.286\delta + 0.402\chi_\phi + 0.652\chi_\phi^2 \quad (53)$$

$$d\phi_{330} = 2.759 + 0.406\delta - 0.055\chi_\phi \quad (54)$$

$$d\phi_{440} = 4245.459\eta^{5.646} - 0.365\chi_+ \quad (55)$$

where $\chi_\phi = 1/2[\delta\chi_+/(1-2\eta) + \chi_-]$. Here all the differentials corresponds to $d\phi_{lmn} = \frac{m}{2}\phi_{22n} - \phi_{lmn}$. We chose to have ϕ_{220} and A_{220} as free parameters. An interesting aspect to underline is the choice of neglecting overtones. It has been observed that the use of the first overtone $n = 1$ may allow to reproduce the signal solving the problem of choosing the initial time of the ringdown [27] right after the peak of the signal. Otherwise the hypothesis of neglecting overtones is efficient when choosing a later starting time. In this project we calibrated the amplitudes and phases over a later starting time so we cannot consider the overtone $n = 1$. The number of parameters has now decreased with the addition of equations (53-55) and it can still be modified. When we introduced the vector Θ it also contained the QNMs for the modes we considered. These QNMs can be interpolated for a given final mass and spin since they come from a numerical solution of the Teukolsky equations [8]. For this interpolation we refer to Emanuele Berti's web-page which contains the tabulated values of QNMs for a given mass and spin.

In the end, we are going to fix some extrinsic parameters like the position in the sky (θ, ψ) and the inclination angle $\iota = \pi/3$ which, as we saw in the previous section, influence the value of the pattern functions and the spherical harmonics. We finally obtain a vector of 7 free parameters $\Theta = \{M, \chi_f, \chi_1, \chi_2, q, A_{220}, \phi_{220}\}$ where we chose to maintain the χ_1, χ_2 parametrization which demonstrated to be more convenient in the sampling process. Also, we fixed the distances D_L for each event when generating the entire population but there is no need to measure it explicitly. In fact, we reabsorbed the distance in the expression of A_{220} and did not consider it as a free parameter to measure. It means that we do not measure the distances of event starting from the signal.

3.2.2 Single event implementation

Before introducing the analysis on the population we make some useful insights on the implementation of the test over a single event. We take as an example a simulated ringdown from a BH: the properties we assumed are reported in Table 1 where we decided to fix the position in the sky, the mass and the spins of the progenitors. Starting from the features in Table 1 we observe how to build the vector $\Theta_0 = \{M_f, \chi_f^{meas}, q, \chi_1, \chi_2, A_{220}, \phi_{220}\}$ in order to perform the test. Assuming $\phi_{220} = 0$ we just have to calculate the amplitude of the dominant mode by the GR based equation

$$A_{220} = 0.86\eta \quad (56)$$

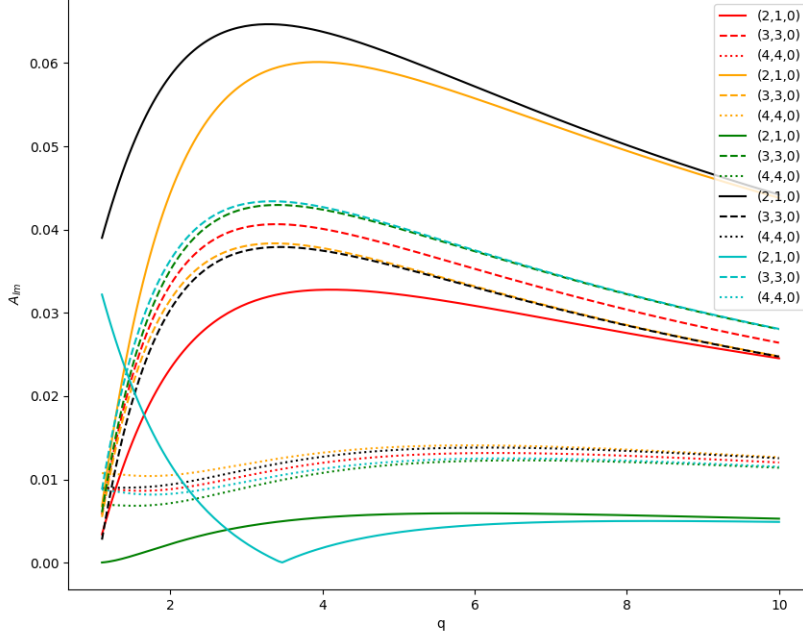


Figure 2: This figure shows the different excitation for modes as a function of progenitors mass ratio and spins. Every line style marks for a different mode and every color for a different choice of spins: $\chi_1, \chi_2 = 0$ (*red*), $\chi_1, \chi_2 = -0.8$ (*orange*), $\chi_1, \chi_2 = 0.8$ (*green*), $\chi_1, \chi_2 = (-0.8, 0.8)$ (*black*) and in the end $\chi_1, \chi_2 = (0.8, -0.8)$ (*cyan*). We chose these values of the spins in order to have a qualitative hint of the different behavior for null, high, low and different spins. At first, we can see that most of the subdominant modes grow with mass ratio and assess to a constant value for high q . The (4,4,0) mode is generally less excited than the (3,3,0) for all spin conditions. The mode (2,1,0) is the most influenced by spins conditions. We see that for null initial spins it excites like the (3,3,0) while for spins which differ from each other it is the most (black) or least (cyan) excited. There is also a difference based on which BH in the binary has a certain spin (cyan, black).

Table 1	
M	$70M_{\odot}$
χ_1	0.
χ_2	0.
q	1.2
RA	1.16
DEC	-1.19
ϕ	1.12
D_L	450 Mpc

Tabella 1: In this table we chose to report some quantities we fixed in the exploration of a single event analysis of the test. The table shows extrinsic properties like the position in the sky.

from [12] and the final spin from equation

$$\chi_f = \chi_t + \eta(L_{\text{ISCO}}(\chi_{eff}) - 2\chi_t E_{\text{ISCO}}(\chi_{eff}) - 1) \sum_{ij} k_{ij} \eta^{(2+i)} \chi_{eff}^j \quad (57)$$

$$k = \begin{pmatrix} -3.82 & -1.2019 & -1.20764 \\ 3.79245 & 1.18385 & 4.90494 \end{pmatrix} \quad (58)$$

$$\chi_{eff} = \chi_t + 0.41616\eta(\chi_1 + \chi_2) \quad (59)$$

$$\chi_t = \frac{(\chi_1 m_1^2 + \chi_2 m_2^2)}{(m_1 + m_2)^2} \quad (60)$$

where L_{ISCO} and E_{ISCO} are respectively the specific (dimensionless) angular momentum and energy for a test particle at the innermost stable circular orbit (ISCO) of a Kerr BH with spin parameters χ_1, χ_2 [28]. We point out that these equations, which determine the final spin of a BH obtained by a merge of a BBH, has been calculated assuming GR to be true; this aspect will play an important role since it means we are injecting GR.

Once we obtain this vector, the strategy we follow consists in sampling from a probability distribution of the vector of parameters $(\Theta - \Theta_0)$ through Bayesian inference. For the implementation on the single event we need to determine a variable which quantify possible deviations from GR when calculating the spin. $\{q, \chi_1, \chi_2\}$ can be used to determine a distribution of the spin (χ_f^{infer}) to compare with the sampled distribution of the same variable (χ_f^{meas}). We define

$$\epsilon = \chi_f^{(meas)} - \chi_f^{(infer)} \quad (61)$$

as the variable that quantifies the deviations from GR. ϵ may be zero when GR is true and may not be zero otherwise. Since both spins have been determined assuming GR we expect to have ϵ distributed around zero with some precision. In order to sample the properties we are interested in, we ask which is the probability distribution of the vector $(\Theta - \Theta_0)$. Equation (26) express the relationship between the signal and the strain through noise. We assume to have Gaussian

noise $n(t) = s(t) - h(t)$. In the limit of *linear signal approximation* (LSA), which corresponds to an 'high' value of the SNR, it has been demonstrated [29] that we can determine the probability distribution of a vector $(\Theta - \bar{\Theta})$, which indicates the deviation of the parameters vector respect to the mean value, where

$$\bar{\Theta} = \Theta_0 + \Theta_m \quad (62)$$

is

$$p(s|\Theta) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} \exp\left\{-\frac{1}{2}(\Theta - \bar{\Theta})^T \Sigma^{-1} (\Theta - \bar{\Theta})\right\}. \quad (63)$$

k represents the length of the parameters vector while

$$\langle \Theta_m \rangle = 0 \quad (64)$$

$$\Sigma^{-1} = \Gamma = \left\langle \frac{\partial h_i}{\partial \Theta_i} \left| \frac{\partial h_j}{\partial \Theta_j} \right. \right\rangle. \quad (65)$$

The sampling process needs a wider explanation. First of all the vector Θ_0 corresponds to the true value of the parameters vector or the value the nature gave to that particular event. The features of this vector have been reported in Table 2. In an actual experiment, this value is unreachable to us because of noise. Although we cannot avoid the presence of noise, noise itself can be modelled as Gaussian with zero mean, equation (64), and covariance matrix Σ in equation (65) [29]. In our case, since the event is simulated, we assume to know the vector Θ_0 and we add Gaussian noise after determining the covariance matrix in equation (65). This process is called *noise realization* and it allows the analysis to be more realistic, adding a certain degree of uncertainty in our test. Noise realization allow us to build the vector in equation (62) which we use as the new true value of the distribution. The covariance matrix has been determined as the inverse of the *Fisher matrix* Γ which depends on the derivative of the strain respect to the parameters. This means that the calculation of the covariance matrix requires the strain h obtained by equation (24) and (25). Also, the product between the derivatives in equation (65) is a scalar product weighted on the PSD of Einstein Telescope. In this way the strain is evaluated according to the precision the interferometer may reach at that frequency.

We point out that the LSA is a good approximation when the value of SNR are high but what we really mean is that there is no threshold for the value of SNR in order to make this assumption to work. The highest the SNR is, the more efficient LSA would be since the higher terms of the probability in equation (63) are $\mathcal{O}(1/\rho)$. Another important issue about the probability distribution related to the vector Θ is the choice of the prior. We refer to an uniform prior for both initial and final spins $\chi \in [-0.99, 0.99]$ and mass ratio $q \in [1.1, \infty]$. Indeed, the presence of the prior influences the process of sampling since it adds some hard bounds on spins and mass ratio. We will discuss the importance of the prior in the population section. Figure 3 represents the implementation of the test for an event affected with noise realization where the features of the event which have been used to produce the ringdown signal from where we inferred the properties are in Table 2.

Θ_0						
$M(M_\odot)$	χ_f	χ_1	χ_2	q	A_{220}	ϕ_{220}
70	0.68	0	0	1.2	$1.59 \cdot 10^{-21}$	0

Tabella 2: Features of a simulated vector Θ_0 . This vector contains the properties of both a BBH and the remnant BH it produces according to GR. This vector has 7 parameters and it has been used to generate the signal.

We obtained a distribution that can be approximated with a Gaussian with mean $\mu = 0.009$ and $\sigma = 0.12$. We performed a *Kolmogorov-Smirnov test* [30] in order to validate the agreement between the two curves and obtained a *p-value* $p > 0.99$. Previously we introduced that in a simulated event like this both of the spin χ_f^{infer} and χ_f^{meas} depend on GR. This consideration is coherent with Figure 3 since ϵ is approximately distributed around the null value. In this fashion, the test reproduces an agreement with GR on a single event and the σ we found determines an indicator of the impact of noise and statistical uncertainties that may appear. The fact that ϵ is distributed with a finite standard deviation is an important indicator of the attention one has to pay when testing GR. In fact, not every detected deviation represents a deviation from GR but these have to be validated, for example, considering a sample of events. We also point out that the deviations this test is able to detect are *model agnostic*. That means that the deviations do not address a specific GR theory but produce an independent result. This aspect may be an advantage since looking for specific deviations may be prevented by the lack of well defined alternative gravity theories.

3.2.3 Mass ratio and Distance

In this section we explore some interesting insights for events with different mass ratio and distances. First of all we set an array of mass ratio $q = \{1.2, 1.4, 2, 3, 4, 5, 6, 7, 8\}$ to determine how the SNR of the event change for different mass ratios. The events will have the same features as in Table 1 except from mass ratio. We evaluate the SNR following equations (24) and (25) for the strains and (46) for the proper quantity. The results can be observed in Figure 4 where we analyzed the behavior of the SNR fixing $\chi_2 = 0$ and $\chi_1 \in [-0.99, 0.99]$ at first and then the other way around. The interesting aspect about Figure 4 is the growth of the SNR from negative to positive spins and the decrease as a function of mass ratio. This result is coherent since we expect the excitation modes behave like $A_{lmn} \sim 1/q$ for higher mass ratio.

The realization of the test respect to the array of mass ratio brings 9 distribution for the 9 different q . We proceed by building the 9 parameter vectors Θ_i starting from features of Table 1 and using equation (56) and (57) in order to determine the dominant mode amplitudes and the final spins. We sample the parameter vectors in the same way we did for the single event and generate 9 different distributions for ϵ . We can see in Figure 5 the differences in the distributions respect to the mass ratio values. The spread of the distributions

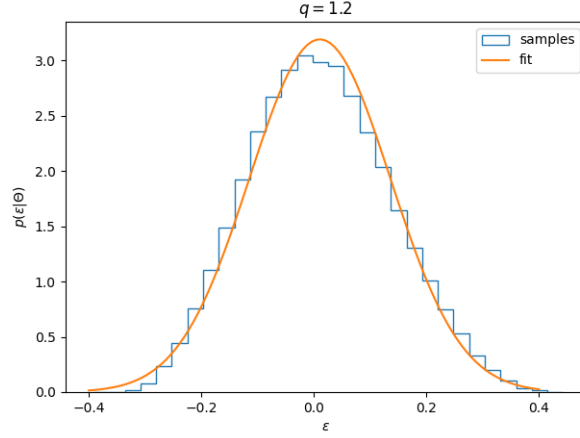


Figura 3: This figure shows the distribution of the variable ϵ which quantify the deviation from GR in the MRC test in a single event. In this case we chose an event with mass ratio $q = 1.2$. As we can see from the figure, ϵ is normally distributed around the null value as we expected since we are injecting all quantities based on GR. The standard deviation we obtained represents a quantitative indicator of how noise affect our measures.

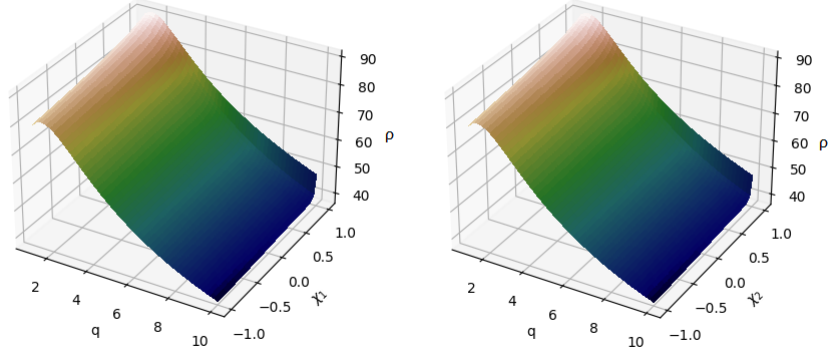


Figura 4: This figures show the behavior of the SNR as a function of mass ratio and initial spins of the progenitors. As we can see the SNR increases for positive spins and decrease for higher mass ratio. This is a consequence of the excitation modes which indeed depend from progenitors properties.

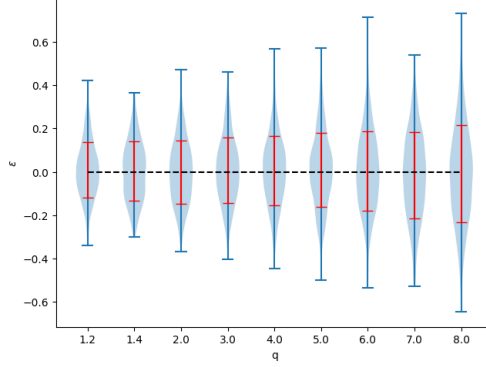


Figura 5: This figure shows the behavior of the probability distribution of the test variable ϵ for an array of mass ratio. The probability distribution gets wider with mass ratio due to the impact it has on excitation of the subdominant modes. The red vertical lines constrain the 90% of the credible interval of the distributions.

gets wider with q , so the uncertainties for the test variable increase. This result can be explained with the description we gave of the amplitudes: the subdominant modes are less excited as the mass ratio increases and this phenomenon produces a decrease in the precision of the probability distribution of ϵ .

We now explore the same distribution but now fixing the SNR and allowing the distance to vary. We ask which distance these 9 events should have to be detected at $\rho = 100$. Figure 6 shows that fixing the SNR bring all probability distributions to have similar width. The reason why this happen is that minor distances approximately reduce the problem of non excited subdominant modes since these depend on the inverse of the distance as in equation (1).

3.2.4 The impact of measuring spin

We are now interested in lowering the number of measured parameters and observe the effect this procedure has on the implementation of the test. In particular, we decide to exclude the initial spins from the vector Θ to have 5 parameters to measure when sampling the distribution. We maintain the values of Θ_0 in Table 2 but now we do not include the initial spins χ_1 and χ_2 in the parameters estimation, assuming we have a prior knowledge of these quantities. This assumption influences the calculation of the fisher matrices, which are now based on 5 parameters, and the sampling process itself which does not include the initial spins. It means also that the prior distribution for initial spins has not to be included, allowing a less restrictive sampling for the vector Θ . We follow the same process of the previous section, sampling from the 5 parameters distribution the vector $\Theta = \{M_f, \chi_f, q, A_{220}, \phi_{220}\}$ with noise realization, calculating the inferred spin with equation (57) and the test variable with equation (61). The results are represented in Figure 7. We fitted the

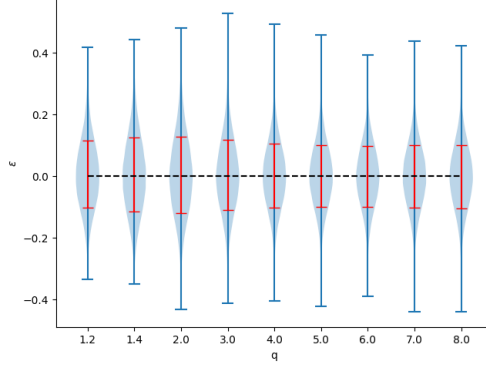


Figura 6: This figure shows the behavior of the probability distribution of the test variable ϵ for an array of mass ratio fixing the SNR $\rho = 100$. This has been possible by letting the distance vary for the events. The probability distributions have approximately similar width compared to previous figure where the effects of different mass ratio influenced the excitation of the subdominant modes. This means that fixing the SNR reduces the effect of the excitation modes on the measures. The red vertical lines constrain the 90% of the credible interval of the distributions.

new distribution with a Gaussian fit obtaining $\mu_{known} = 0.0005$ and $\sigma_{known} = 0.0024$. By these Gaussian parameters we can notice that the magnitude of the impact of a prior knowledge for initials spin translates in a more accurate final distribution for ϵ . Both mean value and standard deviation decrease by one order of magnitude.

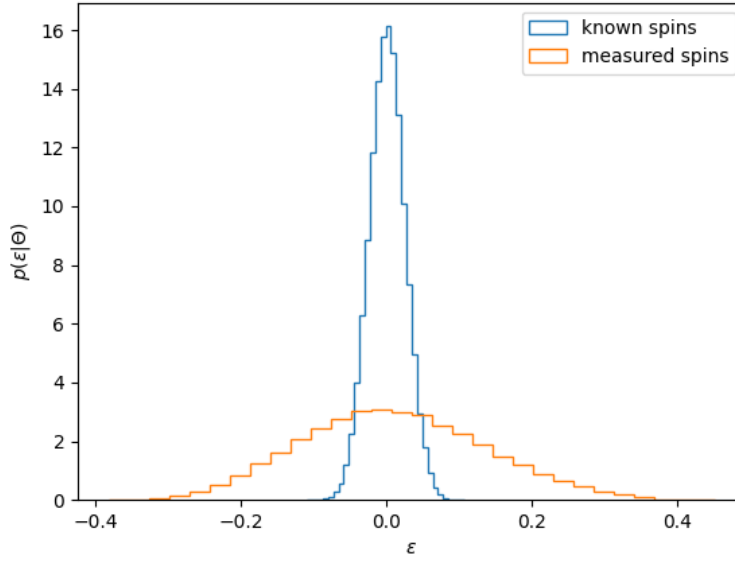


Figura 7: Comparison between the two test variable distribution in the measured initial spin case and in the known spins case. As we can notice, in the second situation the distribution is thinner than the first case. This result represents the gain we have in accuracy when considering less parameters in the sampling process. What happens is that the uncertainties obtained by the fisher matrices are lower because we neglected the influence of the measure of the initial spins. We found $\mu_{known} = 0.0005$ and $\sigma_{known} = 0.0024$ compared to $\mu_{meas} = 0.009$ and $\sigma_{meas} = 0.12$ of the first case.

4 Chapter III

4.1 Implementation on the population

We now describe the implementation of the test on $N = 500$ BBHs. We simulated the population assuming uniform masses $m_1, m_2 \in [30, 180]M_\odot$ for binaries, where $m_1 > m_2$, so that the mass ratio maximum value is $q = 6$. Also the population is assumed to be uniformly distributed in volume in a range $D_L \in [100, 5000]$ Mpc [31]. We made a series of assumption in order to use the parametrization we introduced in the previous section. The sky position have been set to be equal for each binary and the inclination angle $\iota = \pi/3$. The position in the sky as right ascension and declination has been reported in Table 1. We consider the modes (2,2,0), (2,1,0), (3,3,0) and (4,4,0) and fix the initial spins of the binary $\chi_1, \chi_2 = 0$. We also fix $\phi_{220} = 0$ and determine A_{220} with equation (56), the final spin with equation (57) and the final mass for each event. The final mass has been computed by the following equation

$$M_f = M_T - M_{rad} \quad (66)$$

$$M_{rad} = E_{rad}M_T \quad (67)$$

where E_{rad} is the energy radiated by a BBH in gravitational waves and M_{rad} is the mass loss [32]. This equation contains the information about how much mass a BBH loses in the form of gravitational waves according to GR. We already explored the implications of these assumptions and we now know that with this information we are able to define Θ_0 for each event. Then, we generate the waveforms through equations (24) and (25) and calculate the SNR with equation (46) for the entire population. Since the SNR plays a fundamental role in the sampling process¹ we established $\rho < 12$ as threshold to discard events, bringing the number of events from $N = 500$ to $N = 488$. After this selection process we determined the Fisher matrices in equation (65) with respect to the 7 parameters we have as true value². Then, we add the noise realization sampling the parameters Θ in equation (62) for the whole population from the probability distribution with covariance matrix deduced by fisher matrices.

An important aspect to underline is the choice of uniform priors for mass ratio, final spin and initial spin which add some hard constraints both on the noise realization than the sampling process, where we sampled $n = 1000$ points for each event. We report in Figure 8 the mean values $\bar{\epsilon}_i$ of the probability distribution of the population where we indicates ϵ_i as the test variable for the i -th event. This figure shows how distant the mean value of an event is from zero depending on its mass ratio and its SNR. All SNR have been represented in a logarithmic scale and Figure 8 shows that most events with lower SNR

¹we remind that the LSA is more suitable for higher SNR

²An important aspect to point out is the inefficiency of the equations (53,55) for $q = 1$. For this particular value of the mass ratio the calculation of the fisher matrix result to be impossible since the quantity $\chi_\phi \sim 1/(1-q)$ is undefined. This situation represents a limit where the fits of the phases ϕ_{lmn} are not suitable. This means that BBH with the same masses cannot follow our parametrization

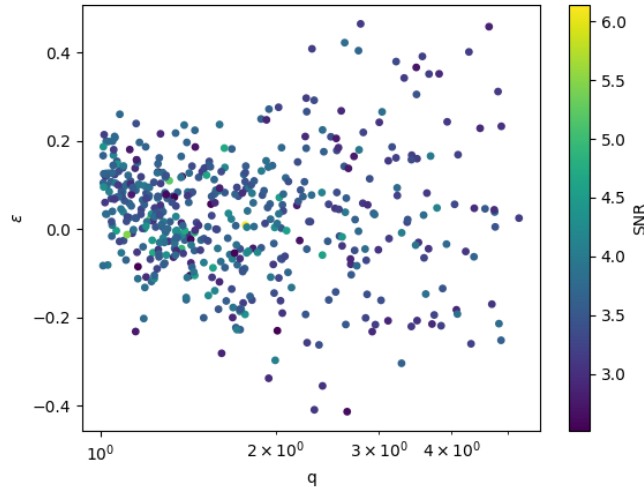


Figura 8: In this figure it is represented the means of the ϵ_i probability distribution for the whole population as a function of their mass ratio. The SNR has been added as a possible indicator of the detectability of the events. SNR have been represented in a logarithmic scale.

appears to be further from zero. Also, we can see that events with a smaller mass ratio tends to be slightly closer to zero compared to events with higher mass ratio. This again can be justified by the relationship between mass ratio and excitation of the modes.

From Figure 8 it is obvious that the mean values of some events are not zero. Do these events manifest deviations of GR or a failure of the test? In order to answer this question we have to introduce an indicator which is able to tell if the zero value have some sort of relevance in the probability distribution of that event. For this reason the *general relativity quantile* (Q_{GR}) represents the perfect metric to evaluate if a certain event manifests a deviation from GR or not. The Q_{GR} of a specific value ϵ_i^0 indicates the probability of measuring $\epsilon_i < \epsilon_i^0$. In our case $\epsilon^0 = 0$ for each event. We extrapolated the probability distribution function from the samples of ϵ_i and calculate Q_{GR} for each event in a frequentist fashion, considering the ratio between the number of times $p_i(\epsilon_j) < p_i(0)$ and the number of samples for the singular event, where the subscript j stands for the different values of ϵ for a single event and p_i stands for probability distribution for different events. The distribution of Q_{GR} across the whole population is an useful indicator to address the question about if some events actually violate GR. We established a threshold for this quantity and if $Q_{GR}^i > 0.99$, so the probability to find zero applying the test to the i -th event is higher than 99%, we will consider that measure as a violation of GR. Since this remains a simulation and we are injecting GR in the data,

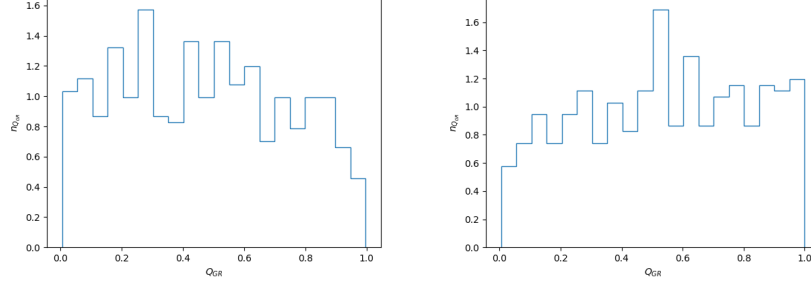


Figure 9: Q_{GR} distribution of the BBH population. This probability distribution represents an indicator of how many events in the population violates the test. Since the injected method is based on GR one should expect to find no violations in these settings. For this reason we may refer to the number of violations in these settings as a false positive detection. This may be an indicator of the probability our test has to fail when implementing it on a population due to noise. Right panel indicated the Q_{GR} distribution of the 7 parameters estimation problem while left panel represents the same distribution for the 5 parameters problem. The percentage of $Q_{GR} > 0.99$ which may represent a violation of GR is respectively 0.6% and 1.2%.

if there are some events which according to this metric violate GR they might constitute a possible *false positive* event, where GR seems to manifest violations because of the presence of noise. The number of $Q_{GR} > 0.99$ may represent an indicator of the number of false positive this tests provides. Otherwise, having a $Q_{GR} > 0.99$ does not mean that a false positive is present since in a real situation we do not know if the reason behind this value is an actual deviation or noise, therefore a real situation may need a way to distinct the nature of an event with $Q_{GR} > 0.99$. Figure 9 shows the Q_{GR} distribution of the population of BBH we just analyzed with $p(Q_{GR} > 0.99) \sim 0.6\%$. So, there 0.6% of probability that this settings for the test provide an event which may violates GR and this aspect have to be taken into account in a realistic scenario. We tried the same implementation with zero noise³ finding no events with $Q_{GR} > 0.99$ like it is showed in Figure 10. This means in these settings we have no violations of GR without noise realization meaning that the percentage of events with $Q_{GR} > 0.99$ in the first case has been produced by noise.

We are interested now in addressing a second question: can we make considerations about the behavior of the whole population instead of the single event? To answer this question, we introduce the concept of *stacking*, which is a process based on the idea of generating a variable ϵ that indicates if the entire population manifest deviation from GR instead of looking for violation for the single event. Performing stacking requires the definition of a domain D of the

³The zero noise case has been obtained by using Θ_0 as mean value of the probability distribution in equation 63

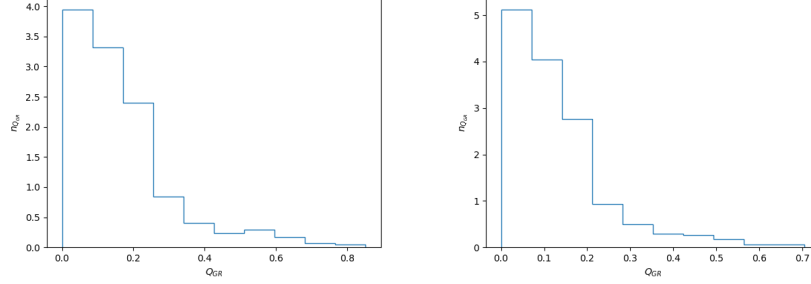


Figura 10: Q_{GR} distribution of the population for the zero noise case. As shown in this figure, there is no event with $Q_{GR} > 0.99$ which represent possible violations of GR in both the 7 parameters analysis (left panel) and the 5 parameters one (right panel). This plots indicates that the presence of noise implies the presence of false positives which may be confused with events that violate GR.

variable ϵ_i and refer to the total posterior distribution of the population in this domain

$$p(\epsilon|d) = \frac{1}{p(\epsilon)^{l-1}} \prod_i p(\epsilon_i|d). \quad (68)$$

where d represents the data. To explain where equation (68) comes from we point out that the sampling process returns a posterior distribution for the single event. We determine the total posterior distribution by equation

$$p(\epsilon|d) = \prod_i p(\epsilon_i|d) = \prod_i [p(d|\epsilon_i)p(\epsilon_i)] \quad (69)$$

where all the priors are the same. Since the real total posterior distribution takes into account only one prior it can be calculated by dividing the previous equation with the $N - 1$ power of the prior.

Equation (68) refers to the probability of having a specific value $\epsilon_i \in D$ for all the events at the same time. The domain D has been set assuming the distribution of the new stacked variable ϵ to be a gaussian with mean value μ_{eff} and standard deviation σ_{eff}

$$\begin{aligned} \mu_{eff} &= \frac{\sum_j w_j [\frac{1}{N} \sum_i \epsilon_{ij}]}{\sum_j w_j} \\ \sigma_{eff} &= \frac{1}{\sum_j w_j^{1/2}} \\ w_j &= \frac{1}{\sigma_j^2} \end{aligned}$$

where the subscript j represents the event and i stands for the number of samples of the same event. Than D has been considered as a specific distance from the

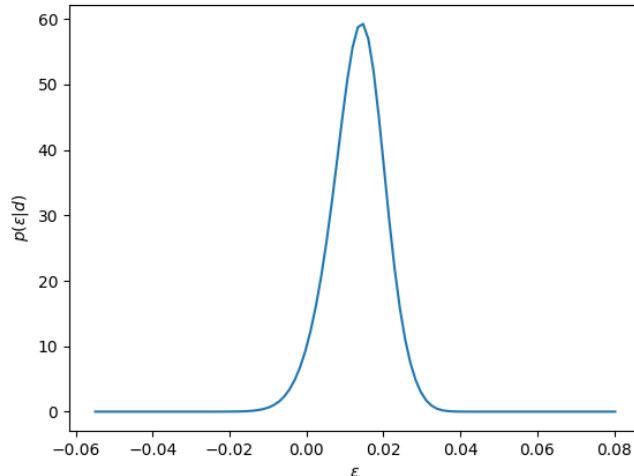


Figura 11: This figure shows the probability distribution of the stacked population. We generated a variable ϵ to represent the behavior of the population as a whole instead of analyzing the single events. This distribution can be fitted with a gaussian distribution $\mathcal{N}(\mu_{stack}, \sigma_{stack})$ where $\mu_{stack} = 0.023$ and $\sigma_{stack} = 0.008$. In this settings we managed to have a 10% level of accuracy respect of the mean value.

mean value

$$D = [\mu_{eff} - 10\sigma_{eff}, \mu_{eff} + 10\sigma_{eff}]. \quad (70)$$

In addition , in equation (68) the role of the prior $p(\epsilon)$ has to be considered. This prior has been determined by calculating the distribution of the variable ϵ_i considering uniform distributions for mass ratio $q \in [1.1, \infty]$ and initial spins $\chi_1, \chi_2 \in [-0.99, 0.99]$. That means we assumed to perform the test with no information about mass ratio and spins as we can see in Figure 12.

In the end a normalization factor has been added. The result is the probability distribution showed in Figure 11. In this probability distribution the value $\epsilon = 0$ of the stacked variable has probability that it is not part of the 95% credible interval, which we establish to be the requisite to not be considered a deviation. The probability distribution in Figure 11 can be fitted by a gaussian $\mathcal{N}(\mu_{stack}, \sigma_{stack})$ where $\mu_{stack} = 0.023$ and $\sigma_{stack} = 0.008$, making the $\epsilon = 0$ value to be at 3σ from the mean value. We repeated this simulation for several samples obtained by the same population with coherent results.

In the end we compare these results with the 5 parameters case applied on the same population. In this last analysis we exclude the initial spins from the parametrization and applying the same process we did before. As already introduced, the difference between the two cases would be more accurate sam-

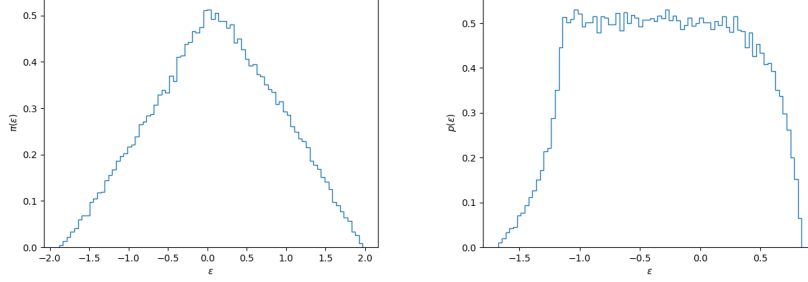


Figure 12: Prior distribution of the test variable ϵ in the 7 (left panel) and 5 (right panel) parameters estimation process. These prior distributions have been determined by implementing the test for assuming uniform distribution for initial spins and mass ratio.

pling due to the exclusion of the uncertainties of the initial spins. In addition, the prior on ϵ will be almost uniform depending only on the mass ratio prior distribution (Figure 12). Figure 13 shows the two stacked distribution in the two different cases. The 5 parameters distribution can be fitted by a gaussian $\mathcal{N}(\mu'_{stack}, \sigma'_{stack})$ with $\mu'_{stack} = 0.005$, $\sigma'_{stack} = 0.002$. Comparing the two distributions we can notice that known spins have an impact on the width of the distribution of ϵ which is tighter than the previous case. Also, the mean value results to be one order of magnitude lower compared to when we measured spins, but is there an effective advantage overall? Despite we are able to measure the ϵ more accurately we can see from Figure 13 that the probability associated with the null value is almost the same. So, the accuracy we gained is also combined with a decrease of the standard deviation. In fact, $p(\epsilon|d) = 0$ happens to be at $2.5\sigma_{known}$ in the probability distribution with known spins compared to a distance of $3\sigma_{meas}$ of the first case. Despite the distance from the mean value has decreased the null value is still outside the 95% credible interval.

In addition, the percentage error for the know spins distribution appears to be $\delta = 100\%$ of the measure compared to the $\delta = 10\%$ of the previous case. Besides, the analysis of the Q_{GR} distribution for the known spins case reports a $p(Q_{GR} > 0.99) = 0.012$ compared to the previous $p(Q_{GR} > 0.99) = 0.006$ of the measured case. Since the zero noise case has been computed also for the known spin case finding again no events with $Q_{GR} > 0.99$, we can interpret the result by saying that the known spin case may be more sensitive to noise. We point out that this situation where these distributions do not centre in $\epsilon = 0$ and the possible disadvantage the prior knowledge of spins brings, may need a deeper investigation rather than concluding a grade of inefficiency of the test. In fact, running the simulation with no noise realization leads to the results in Table 3 in terms of stacking variable distribution. We see that the features of the probability distributions in the zero noise setting correspond to a slight improvement manifested in the decrease of the distance d between the mean

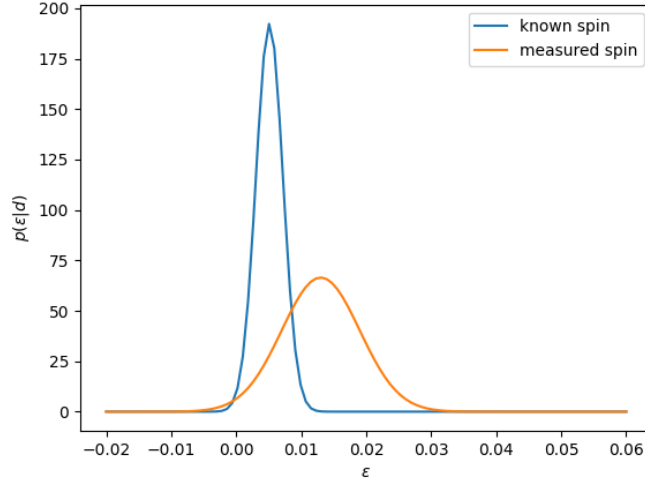


Figura 13: Comparison between stacked distribution of the test variable in case of measured spins and known spins. As we can see from this plot, the standard deviation in the known spin setting is lower than the measured spin case; therefore the accuracy of the measure increases in the 5 parameters problem. The value $\epsilon = 0$ happens to be at $3\sigma_{meas}$ compared to the second case $2.5\sigma_{known}$.

value of the distribution and the null value. Therefore we may attribute the residual deviation we get to a systematic error instead of noise. We will debate the possible nature and the presence of this systematic error in the next sections.

4.1.1 Injected deviations

In the previous section we showed that the performances of the MRC test for a population of BBH can give an estimate of the absence of a violation of GR with a value $p(\epsilon = 0|d)$ at $3\sigma_{meas}$, in the measured spin case, and at $2.5\sigma_{known}$, in the known spins case, from the mean values. It means that the null value for the stacked distribution indicates that the absence of violations has a probability lower than 1%. We calculated the test variable in equation (61) by measuring the spin of the remnant black hole and obtaining the same quantity from measurements of the mass ratio and initial spin of the BBH by equation (57). All these measures have been sampled from the ringdown signal and the final spin values we obtained for all the N events with both methods are based on GR. For this reason, idealistically, we expect to find no deviations in the population. In this section we change approach in order to verify if the test is able to detect injected violations of GR. We will add two types of violations: sistematic and gaussian, following the suggestions in 3.1.1. But how can we inject deviations from GR in order to observe the performance of the test? The answer is that

Measured Spins				
	μ	σ	d	$\%Q_{GR} > 0.99$
NR	0.023	0.008	3	0.6
WNR	0.015	0.007	2	0
Known Spins				
	μ	σ	d	$\%Q_{GR} > 0.99$
NR	0.005	0.002	2.5	1.2
WNR	0.004	0.002	2	0

Tabella 3: This table contains the mean values and the standard deviations for the 7 and 5 parameters implementation of the test. The column d represents the distance in units of σ between $\epsilon = 0$ and the mean value of the corresponding distribution. In this table have been reported both measures of the analysis which includes noise realization (NR) and the one who does not (WNR). Despite there is no advantage in the accuracy of the measure We can observe that without noise realization the distance d decreases in both situations.

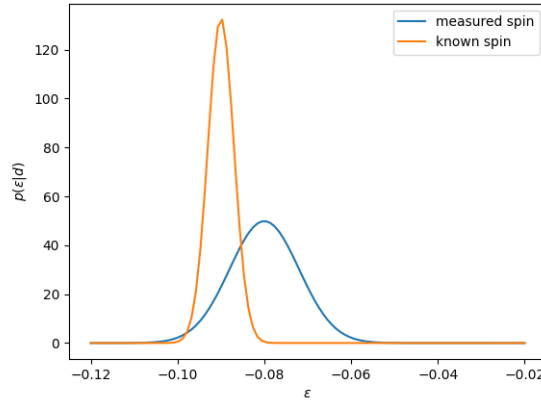


Figura 14: This image shows the stacked distribution of the ϵ variable in both measured spins and known spins cases. As we can see from this image the two distributions differ from each other by width and mean value. In particular, the mean value of the measured spins distribution happens to be less than the injected because of the influence of the injections in the sampling process.

the violation has been added when we determine the true values for each BBH. In order to sample the probability distribution of the single BBH we had to produce a vector Θ_0 based on the population we generated and equations based on GR. This time, instead of following equation (57) to determine the final spin of the remnant BH we use the same equation with the addition of a constant $c_0 = -0.1$ at first and then a gaussian distributed value with $\mathcal{N}(\mu_{dev}, \sigma_{dev})$ where $\mu_{dev} = 0$ and $\sigma_{dev} = 0.01$ in a second case. This approach should allow the spin of the remnant black hole we sample from the distribution to not follow GR, while the ones we calculate using mass ratio and initial spins may still stick with Einstein's theory. Of course we expect that the sampled distribution of mass ratio, initial spins and final spin will be slightly influenced by these injections. For the systematic deviation we repeat the process we explained in the previous section in both measured and known initial spins cases, obtaining Figure 14, where it is clear that both distribution manifest the presence of a deviation from the null value. These distributions can be respectively fitted by gaussian functions which mean value and standard deviation are reported in Table 3. There are several differences between the two parametrizations: first of all, knowing the initial spins forces the distribution to center closer to the value we expect compared to the measured spins case. The explanation for this phenomenon may be the fact that injecting deviations in the spin value influences the sampling process for both mass ratio and initial spins. For this reason when we calculate final spins by sampled initial parameters $\{q, \chi_1, \chi_2\}$, these samples are not precisely the ones we would obtain when GR is true. In this fashion the prior knowledge of spin have an non negligible impact since we are able to measure the true value of the deviation more accurately of one order of magnitude. As we expected from the analysis of the previous paragraph, standard deviations of the two distributions are different with $\sigma_{meas} > \sigma_{known}$. Figure 14 shows that the population we analyzed in this case manifest a violation of GR collectively. In addition the null value happens to be at $10.5\sigma_{meas}$ from the mean value in the first case while at $28.5\sigma_{known}$ from the mean value in the second case. Referring to the values in Table 4 the distance between the null value and the mean happens to be greater in the known spins situation. Another difference in the application of the two methods may be observed in the Q_{GR} distribution in Figure 15. The Q_{GR} distributions show that in the measured spins case the event with $Q_{GR} > 0.99$ are approximately 0.4% while in the known spins analysis happens to be 9% of the entire population. This means that knowing the initial spins may be an advantage in recognizing events which present violations from GR.

We are now interested in adding a different type of deviation and observe the performance of the test. Therefore, with the same logic which guided us in the systematic violation case, we added a Gaussian perturbation for the final spin of the remnant BHs $\mathcal{N}(\mu_{dev}, \sigma_{dev})$ where $\mu_{dev} = 0$ and $\sigma_{dev} = 0.01$. The assumption of a Gaussian perturbation for the final spin generates a different scenario from the previous situations. The features of the distributions in this last scenario are reported in Table 5 where the $p(\epsilon = 0|d)$ is at $4.5\sigma_{meas}$ for the 7 parameters setting while it is at $3.5\sigma_{known}$ for the 5 parameters analysis.

Measured Spins				
	μ	σ	d	$\%Q_{GR} > 0.99$
NR	-0.008	0.008	10.5	0.8
WNR	-0.09	0.009	10	0.4
Known Spins				
	μ	σ	d	$\%Q_{GR}$
NR	-0.009	0.003	28.5	9
WNR	-0.009	0.003	32	3

Tabella 4: This table contains the mean values and the standard deviations of the two gaussian distribution which represent the stacked population of BBH in case of addition of a constant deviation $c_0 = -0.1$ in the final spin of BHs. We reported the information obtained both with noise realization and zero noise. The percentage of $Q_{GR} > 0.99$ has been reported.

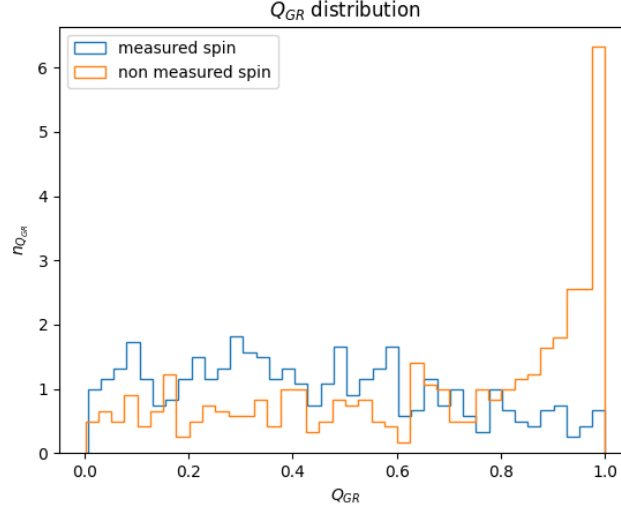


Figura 15: This figure represents the comparison between Q_{GR} distribution of the measured spins and known spins distribution in the $c_0 = -0.1$ case. In particular, knowing the spins produce an higher number of $Q_{GR} > 0.99$ which we associated with events that manifest violations of GR.

Measured Spins				
	μ	σ	d	$\%Q_{GR} > 0.99$
NR	0.036	0.008	4.5	0.6
WNR	0.013	0.009	1.5	0
Known Spins				
	μ	σ	d	$\%Q_{GR}$
NR	0.065	0.002	3.5	2.2
WNR	0.007	0.002	3.5	0

Tabella 5: This table contains the mean values and the standard deviations of the two Gaussian distribution which represent the stacked population of BBH in case of addition of a Gaussian deviation $\mathcal{N}(\mu_{dev}, \sigma_{dev})$ where $\mu_{dev} = 0$ and $\sigma_{dev} = 0.01$ in the final spin of BHs both for measured and known spins. In this table the distance d in units of σ has been added together with the percentage of $Q_{GR} > 0.99$. The probability distributions have been studied with and without noise in order to underline the possible influence of noise.

This indicates a slightly different behavior from the analysis with no deviations since the null value which represents the absence of violations is further from the corresponding mean values. The Q_{GR} distributions for the Gaussian deviation scenario present respectively a 0.6% of probability to find $Q_{GR} > 0.99$ in the measured spins case and 2% probability in the known spins case. Figure 16 shows the comparison between the two stacked distribution for the Gaussian violation phenomenon. An important aspect to underline is the inability of our test to detect Gaussian deviation of an higher order of magnitude for the standard deviation when knowing the initial spins. The reason behind this inability may be purely methodical, since a wider value for the standard deviation and a lower value of the width of the single distributions translate in an higher difficulty in the stacking process. When adding this sort of Gaussian deviation we experimented that there is no common domain to evaluate the distributions and the stacking process returns zero for the whole domain. The zero noise analysis brings a contradictory result when measuring spins: the probability distribution shows a lower distance d from the mean value of the distribution and no events with $Q_{GR} > 0.99$. This may be a tricky result since it seems this kind of setting indicates a better agreement with the hypothesis that GR is true. The MRC test seems to show the presence of Gaussian perturbation in the other situations through the behavior of the stacked distribution. The Q_{GR} distribution cannot underline the presence of possible events which violate the theory in the zero noise case.

We conclude this section pointing out that the injected violations did not follow a specific logical principle. In fact, these injections do not represent a method to constrain the magnitude of the violations. A natural evolution of this work would be to include a model agnostic deviation in the sampling process which may allow to put constraints on which kind of perturbation this test is sensible to.

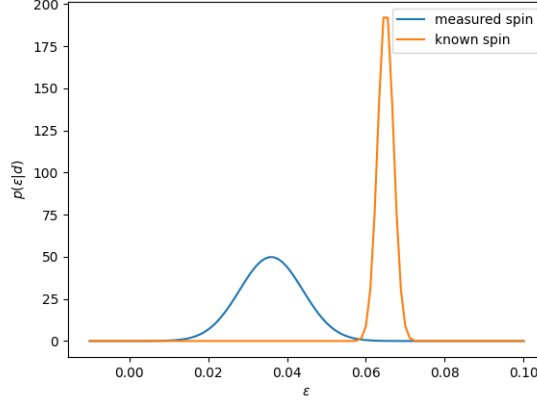


Figure 16: This image shows the comparison between the probability distribution of the test variable with injected Gaussian deviation in the calculation of the spin of the remnant BH in a population of BBHs. Both these distributions manifest the presence of a violation from GR at $4.5\sigma_{meas}$ and $3.5\sigma_{known}$ respectively.

4.2 The role of the prior

In this last paragraph we introduce the possible presence of a systematic bias in our analysis method. We implemented the MRCT for a population of BBHs without considering noise realization and observed contradictory results. We expect to find a better sampled distribution of the test variable in this case while Figure 17 shows that there is a percentage of events with mass ratio $q \sim 1$ which deviates from the null value. This phenomenon happens to be present in both 7 and 5 parameters analysis. Since we assumed uniform priors on progenitors spins and mass ratio it may be possible that these priors are dominating the sampling process. For this reason, events with $q \sim 1$ are being influenced by this phenomenon. The consequence would be an overestimation of the test variable in equation (61) for these events.

The exploration of the possible reasons for the presence of this bias and its nature is an object of inquiry for this project.

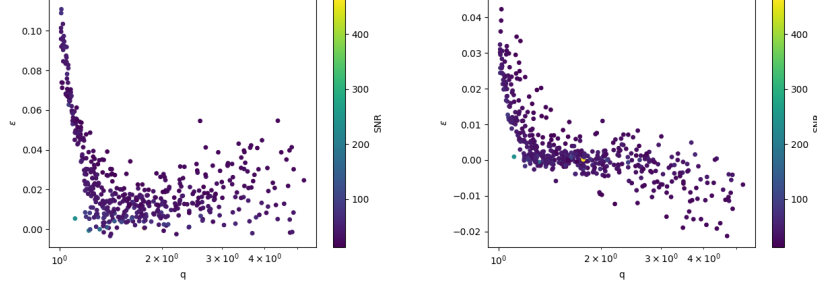


Figura 17: This figure shows the distribution of the mean values of the population of BBHs as a function of mass ratio in the zero noise settings for measured (left panel) and known (right panel) spins. We observe a systematic positive bias for the mean distribution generated by the domination of the prior distribution of mass ratios. In the left plot the higher width of the bias may be explained by the fact that we are considering also the priors of the initial spins

5 Conclusions

This project aimed to present how the Merger Ringdown consistency test may be implemented on a specific population of BBHs. This test uses the ringdown signal provided by a merger of BBHs in order to test General Relativity. The MRC constitutes a null test that needs the definition of a variable ϵ_i which is idealistically zero when GR is true and is not zero when GR has been violated. We build the settings to perform the test at first simulating a single event and then in two different cases for a population of BBHs, including spins in the measure at first and exclude them in a second situation. The single event implementation provided positive feedback with a distribution of the variable ϵ well fitted by a Gaussian distribution $\mathcal{N}(\mu, \sigma)$ where $\mu = 0.009$ and $\sigma = 0.12$, so the value $\epsilon_i = 0$ happens to be in the 95% credible interval of the distribution. We experimented the impact of a prior knowledge of the initial spins of the BBHs on the implementation, noticing that lowering the number of parameters implies an increase of the accuracy of the measure of ϵ . Implementing the test required several assumptions on the ringdown signal which simplify the analysis corresponding to an idealistic scenario. Of course the different modelling of the waveform may help increase the accuracy of this implementation. In addition we do not expect some assumptions we made to be respected in for a real event where initial spins may not be aligned or equal to zero. These aspects have to be taken into account for a more realistic implementation.

The application on a population of binaries generated more controversial results. We generated a population of BBHs in order to produce the ringdown signal we expected if GR was true. Then we inferred the features of the population from this signal and checked the consistency between the inferred final spins of the remnant BHs and the final spins calculated by inferred mass ratios and spins of the progenitors. Then, the population distributions have been stacked by producing a variable ϵ which condensates the behavior of the entire population. The null value of this variable does not fit in the 95% of the total distribution in both measured and known spins settings, despite there is a slight improvement in the second case. The analysis made by excluding the presence of noise highlighted the presence of a systematic bias in the data which generates an overestimation of the ϵ variable due to the presence of priors on mass ratio and spins of the progenitors. This problem can be faced only improving the accuracy of the sampling process by improving the SNR or a new method of analysis has to be taken into consideration. We stress out that our population is simulated and does not follow principles about the origin of BHs in the universe which we expect to be less suitable than our population. For example, we assumed the binaries to be in the same position in the sky which is a strong assumption. In a real situation we might expect the binaries to be in different positions respect to the detector, allowing for a modulation of the signal which may decrease the SNR of the event.

We explored also the behavior of the test for injected spins deviations. We injected a constant violation of GR at first and a Gaussian one in a second case, observing the shift of the distribution of ϵ reported in Table 4 and 5. The

analysis of the presence of injections have been sustained by calculating the Q_{GR} distribution which highlighted the presence of several events which suppose to violate GR in the known spins setting overall.

A possible evolution of this work may include a different approach for the sampling process which might allow to avoid the systematic bias in the data. In addition, different population may be used to check the performance of the test. This kind of analysis can be also implemented considering LISA (*Laser interferometer space antenna*) as an interferometer which will be able to detect supermassive BHs. The Python code for implementing the test on a population of BBHs is available at the following GitHub Repository.

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